Grading Guide
Programming for Economists
Winter ReExam 2025/26

Data Analysis Project

Problem 1: Aggregate Inflation in Denmark

Question 1.1

- Retrieves PRIS113 data via an API call.
- Constructs the CPI index (2020 = 100).
- Computes month-to-month and 12-month inflation rates.
- Produces clear time series plots with labeled axes and legend.
- Provides a short interpretation of when the inflation surge ended.

Question 1.2

- Correctly defines and computes the weights $\kappa(k, \alpha)$ for $\alpha = 0, 1, 2, 3$.
- Computes instantaneous inflation using a rolling window or equivalent method.
- Produces clean comparative plots for each α.
- Discusses how the timing of the inflation surge differs from Question 1.1.

Question 1.3

- Retrieves PRIS111 data for total CPI, CPI excl. energy, and core inflation.
- Correctly computes 12-month inflation rates.
- Plots the three series clearly since 2019.
- Briefly discusses likely drivers of the post-pandemic inflation surge.

Question 1.4

- Retrieves all 4-digit PRIS111 product categories.
- Computes 12-month inflation for each category with `groupby` methods.
- Produces percentile plots (25th, 50th, 75th).
- Produces a histogram of price changes from Aug. 2020 to Aug. 2025.
- Identifies and illustrates the top-10 and bottom-10 categories.
- Gives a short descriptive interpretation of the distributional patterns.

Problem 2: International Comparison

Question 2.1

- Retrieves CPI from Denmark Statistics and HICP from FRED.
- Plots the two indexes on a comparable scale.
- Comments on similarities, differences, and whether the indices are comparable.

Question 2.2

- Retrieves HICP data for Denmark, Austria, Euro Area, and the U.S.
- Plots index levels and 12-month inflation rates since 2019.
- Computes min, max, and mean inflation rates year-by-year.
- Summarises cross-country differences in inflation paths clearly.

Problem 3: Extension

- Extends the analysis with at least one new dataset,
- Produces 1–3 new figures with clear economic relevance,
- Explains the insight yielded by the extension.

Model Analysis Project

Problem 1: Labor Supply

Question 1.1

- Evaluates and plots the utility function over a grid for ℓ_i.
- Evaluates and plots the FOC $\varphi(p_i, c_i, \ell_i)$ over the same grid.
- Finds optimal labor supply using a numerical optimizer.
- Finds optimal labor supply using a root finder.
- Compares the two methods and comments on speed and robustness.
- Compiles results into one clear figure (with subplots allowed).
- Repeats the exercise for the additional ϵ values with minimal new code.

Question 1.2

- Computes the optimal labor supply function $\ell^\star(p_i)$ for $p_i \in [0.5, 3.0]$.
- Correctly handles the minimum feasible hours condition.
- Produces clear plots for both $\zeta = 0.1$ and $\zeta = -0.1$.
- Briefly comments on how the lump-sum component affects labor supply.

Problem 2: Public Good

Question 2.1

- Draws a productivity distribution and computes labor supply for all workers.
- Computes tax revenue $T(\tau, \zeta)$ and the Social Welfare Function.
- Repeats this across the required range of ζ values and τ.
- Plots revenue and SWF clearly.
- Indicates the approximate maximizing point(s).

Question 2.2

- Runs a numerical optimizer to find $(\tau^\star, \zeta^\star)$.
- Checks robustness to starting values.
- Produces local plots of $T(\tau, \zeta)$ and SWF around the optimum.
- Computes and plots the Lorenz curve of consumption at the optimum.

Problem 3: Top Tax

- Evaluates and plots the utility function under the top-tax system.
- Computes and plots the FOC for the pre-kink and post-kink regions.
- Finds optimal labor supply via numerical optimization.
- Computes optimal labor supply via the four-step FOC approach.
- Compares the two methods and comments on speed and differences.
- Repeats the analysis for the additional productivity values with minimal new code.
- Presents results clearly, optionally in a multi-panel figure.

Question 3.2

- Computes $\ell^\star(p_i)$ and c_i for $p_i \in [0.5, 3.0]$.
- Classifies workers into ℓ_b, ℓ_k, and ℓ_a cases.
- Reports the proportions in each region.
- Plots both labor supply and consumption clearly.

Question 3.3

- Computes the SWF under the introduced top tax.
- Computes and plots the Lorenz curve of consumption.
- Explores alternative (ω, κ) values and reports if SWF improves.

Problem 4: Extension

- Implements a meaningful model extension (preferences, heterogeneity, tax rules, etc.).
- Evaluates the extended model correctly.
- Explains the resulting economic insight clearly

Exam Project

Problem 1: Danish Consumption

Question 1.1 (Consumption share of disposable income)

- Retrieves quarterly data from StatBank Denmark table `NKSO2` via an API call (transactions `B6GD` and `P3D`, sector `S1M`).
- Constructs the ratio: consumption / disposable income.
- Performs seasonal adjustment using `statsmodels.tsa.seasonal.STL` (quarterly period=4) for both series.
- Produces a clear plot of the raw ratio and the seasonally adjusted ratio (labeled axes, legend, readable formatting).
- Adds horizontal mean lines for the requested subsamples (pre-2008Q1, 2010Q1–2019Q4, 2021Q1–latest) and clearly indicates which mean corresponds to which subsample.
- Provides a short, correct interpretation of the level changes over the three subsamples.

Question 1.2 (Consumption shares by category)

- Retrieves consumption-by-purpose data from table `NKHC021` (current prices and seasonally adjusted).
- Clean data properly.
- Produces a line plot of total consumption from `NKHC021` and compares it to the consumption series used in Question 1.1 (levels).
- Computes consumption shares for each category (category / total) correctly.
- Plots consumption shares over time with a clear legend (may split into multiple figures/panels for readability).

Question 1.3 (Long-run and post-2019 changes in shares)

- Computes average consumption shares for 1990 and 2025 (the available quarters)
- Computes the change in each category share between the two years.
- Produces a clear bar chart of these changes (sorted bars and labeled categories are rewarded).
- Repeats the change calculation for the post-2019 comparison

Problem 2: Labor supply with non-linear pay schedule

Question 2.1 (Baseline wage schedule)

- Correctly implements the baseline budget and utility components (earnings, consumption, utility) and respects the feasible set $\ell \in [0, \ell_{\max}]$.
- Evaluates and plots $c(\ell)$ and $u(\ell)$ over a grid, with a clearly marked optimum $\ell^\star$.
- Solves for $\ell^\star$ using three distinct numerical methods:
  - grid search,
  - a local optimizer,
  - a root finder applied to the first-order condition.
- The three methods deliver consistent results (up to numerical tolerances), and the solution is interior when expected.
- Briefly comments on method differences (accuracy, speed, robustness).

Question 2.2 (Overtime schedule and multiple local maxima)

- Correctly implements the overtime earnings schedule (kink at $\bar{\ell}$ and higher marginal wage above it).
- Plots $c(\ell)$ and $u(\ell)$ under overtime and demonstrates (visually or numerically) the presence of multiple local maxima.
- Runs the local optimizer from two distinct starting values ($\ell_0 = 0.2$ and $\ell_0 = 1.2$) and explains why the solutions can differ.
- Implements a robust solver that returns the global maximizer (e.g. optimize within each region + check kink points) and documents the approach.
- Presents the final choice $\ell^\star$ clearly and includes a short interpretation.

Problem 3: Labor market simulation

Question 3.1 (Document stationarity)

- Correctly implements the job-finding probability $f_s(d)$ with duration dependence and the lower bound of 0.05.
- Correctly updates employment, unemployment duration (reset to 0 when employed, set to 1 upon separation, increment while unemployed), and human capital dynamics as specified.
- Simulates for $T = 3000$ with $N = 50,000$ and fixed random seed 123.
- Stores only the last 200 periods of the cross section (small panel), and uses it for all stationarity checks.
- Computes and reports unemployment rate and the mean and SD of $\log y$ separately for the first half (-200:-101) and second half (-100:-1) of the stored panel; the two halves are shown to be (almost) identical.
- Compares distributions of $\log y$ and unemployment durations among the unemployed across the two halves (e.g. histograms) and argues stationarity.

Question 3.2 (Compare low- vs high-skilled)

- Produces type-specific summary statistics using the stored panel (pooled over time where appropriate): unemployment rate, mean and SD of $\log y$.
- Computes and reports informative duration statistics by type (e.g. mean duration among unemployed; share unemployed for more than 12 months).
- Presents results in a clear table and provides a short comparison/interpretation.

Question 3.3 (Reasons for inequality / counterfactuals)

- Designs counterfactual simulations that change one type-specific parameter at a time ($\mu_s, a_s, f_{0,s}, \lambda_s$) while keeping the others fixed.
- Re-simulates the model consistently and recomputes the outcome gaps (e.g. Δ unemployment, Δ mean $\log y$, Δ mean duration) for each counterfactual.
- Summarises results in a compact table and clearly identifies which parameters drive the differences.